

Physics · Easier paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ Date _____

121 marks

reading 15 min · writing 150 min

State average on these questions

89 / 121 (73%)

a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright (some exams include third-party material).
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Methods and science working can be typed in plain text, e.g. $v = \sqrt{2gh} = 9.9 \text{ m/s}$.
3. Keep to the real time limits above, and don't look at the guide until you've finished.
4. Save with Ctrl/Cmd + S. Use a PDF app that supports forms: Adobe Acrobat Reader, Chrome, Edge, Firefox, macOS Preview or Xodo.
5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

Every question is a real VCAA exam question, chosen by ranking all the questions from the listed exams by how Victorian students actually scored on them (percentage correct for multiple choice, state average ÷ marks for written questions). This paper takes the easiest questions.

Sources

- [2022 Physics exam \(PDF\)](#) · [examiners' report](#) (*previous study design*)
- [2023 Physics exam \(PDF\)](#) · [examiners' report](#) (*previous study design*)
- [2024 Physics exam \(PDF\)](#) · [examiners' report](#)
- [2025 Physics exam \(PDF\)](#) · [examiners' report](#)

Section A – Multiple-choice questions

Open each question in the official exam PDF (click the blue link), then select A–D. One mark each.

Question 1 · 1 mark

2022 VCAA exam · Section A · Question 1 · [previous study design](#) · [open in the official exam PDF \(p. 2\)](#) ↗

Field direction at the centre of a current loop

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 2 · 1 mark

2022 VCAA exam · Section A · Question 3 · [previous study design](#) · [open in the official exam PDF \(p. 4\)](#) ↗

Charges of particles curving in a magnetic field

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 3 · 1 mark

2022 VCAA exam · Section A · Question 6 · [previous study design](#) · [open in the official exam PDF \(p. 6\)](#) ↗

Mass of a railway truck from a collision

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 4 · 1 mark

2022 VCAA exam · Section A · Question 15 · [previous study design](#) · [open in the official exam PDF \(p. 9\)](#) ↗

Evidence for light as a particle

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 5 · 1 mark

2022 VCAA exam · Section A · Question 18 · [previous study design](#) · [open in the official exam PDF \(p. 10\)](#) ↗

Identifying an inertial frame of reference

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 6 · 1 mark

2022 VCAA exam · Section A · Question 19 · [previous study design](#) · [open in the official exam PDF \(p. 10\)](#) ↗

Lorentz factor of a particle at $2.99 \times 10^8 \text{ m s}^{-1}$

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 7 · 1 mark

2022 VCAA exam · Section A · Question 20 · [previous study design](#) · [open in the official exam PDF \(p. 10\)](#) ↗

What experimental uncertainty represents

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 8 · 1 mark

2023 VCAA exam · Section A · Question 9 · [previous study design](#) · [open in the official exam PDF \(p. 7\)](#) ↗

Banking angle for a velodrome curve

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 9 · 1 mark

2023 VCAA exam · Section A · Question 16 · [previous study design](#) · [open in the official exam PDF \(p. 10\)](#) ↗

Naming the wave behaviour at a barrier

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 10 · 1 mark

2023 VCAA exam · Section A · Question 17 · [previous study design](#) · [open in the official exam PDF \(p. 11\)](#) ↗

Why X-ray and electron diffraction can be compared

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 11 · 1 mark

2024 VCAA exam · Section A · Question 6 · [open in the official exam PDF \(p. 6\)](#) ↗

Field midway between two like poles

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 12 · 1 mark

2024 VCAA exam · Section A · Question 8 · [open in the official exam PDF \(p. 8\)](#) ↗

Force on a current-carrying wire in a field

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 13 · 1 mark

2024 VCAA exam · Section A · Question 11 · [open in the official exam PDF \(p. 9\)](#) ↗

Why electricity is transmitted at high voltage

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 14 · 1 mark

2024 VCAA exam · Section A · Question 20 · [open in the official exam PDF \(p. 15\)](#) ↗

Meaning of “precise” data

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 15 · 1 mark

2025 VCAA exam · Section A · Question 2 · [open in the official exam PDF \(p. 3\)](#) ↗

Direction of acceleration at the top and bottom of a vertical circle

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 16 · 1 mark

2025 VCAA exam · Section A · Question 4 · [open in the official exam PDF \(p. 5\)](#) ↗

Energy changes for a mass oscillating on a spring

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 17 · 1 mark

2025 VCAA exam · Section A · Question 7 · [open in the official exam PDF \(p. 8\)](#) ↗

Static versus non-uniform fields

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 18 · 1 mark

2025 VCAA exam · Section A · Question 15 · [open in the official exam PDF \(p. 13\)](#) ↗

Electrons and X-rays giving matching diffraction patterns

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 19 · 1 mark

2025 VCAA exam · Section A · Question 16 · [open in the official exam PDF \(p. 13\)](#) ↗

Length of a beamline in the protons' frame

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 20 · 1 mark2025 VCAA exam · Section A · Question 18 · [open in the official exam PDF \(p. 14\)](#) ↗

Classifying random and systematic errors

Your answer: ☐ A ☐ B ☐ C ☐ D

Section B – Short-answer questions

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (5 marks)2022 VCAA exam · Section B · Question 1 · [previous study design](#) · [open in the official exam PDF \(p. 12\)](#) ↗

Simple DC motor: current direction in a side, and current from force

a. Direction of current in side KL in Position 1, with justification (2 marks)**b.** Direction of current in KL in Position 3 (1 mark)**c.** Current in the coil from the force on KL (2 marks)

Question 2 (7 marks)2022 VCAA exam · Section B · Question 3 · [previous study design](#) · [open in the official exam PDF \(p. 16\)](#) ↗

Mass spectrometer: accelerating and deflecting ions

a. Increase in kinetic energy of each ion (2 marks)**b.** Show the ion speed is $3.16 \times 10^5 \text{ m s}^{-1}$ (2 marks)

c. Diameter of the semicircular path (3 marks)

Question 3 (7 marks)

2022 VCAA exam · Section B · Question 5 · [previous study design](#) · [open in the official exam PDF \(p. 20\)](#) ↗

Wind generator supplying a factory: losses and transformers

a. Power produced by the generator (1 mark)

b. Can the factory get its 40 kW? Justify with calculations (3 marks)

c. Power delivered with ideal 1:10 and 10:1 transformers (3 marks)

Question 4 (5 marks)

2022 VCAA exam · Section B · Question 6 · [previous study design](#) · [open in the official exam PDF \(p. 22\)](#) ↗

Alternator: field strength from flux, then average EMF

a. Magnetic field strength from the flux (2 marks)

b. Average EMF over the first quarter turn at 2.5 Hz (3 marks)

Question 5 (10 marks)

2022 VCAA exam · Section B · Question 7 · [previous study design](#) · [open in the official exam PDF \(p. 24\)](#) ↗

Trolley on a ramp, then a sticky collision

a.i. Show the component of weight down the ramp is 8.3 N (2 marks)

a.ii. Constant friction force on the ramp (2 marks)

b.i. Speed of the joined trolleys, naming the principle (3 marks)

b.ii. Elastic or inelastic? Show calculations (3 marks)

Question 6 (16 marks)

2022 VCAA exam · Section B · Question 10 · [previous study design](#) · [open in the official exam PDF \(p. 28\)](#) ↗

Tennis-ball launcher investigation: variables, graph, range, air resistance

a. Classify controlled, dependent and independent variables (3 marks)

b. Plot the range–angle data with scales, uncertainty bars and a curve (6 marks)

c. Maximum range and the angle giving it, from the graph (2 marks)

d.i. Theoretical range at 25 m s^{-1} and 30° (2 marks)

d.ii. Evaluate whether air resistance can be ignored (3 marks)

Question 7 (9 marks)

2022 VCAA exam · Section B · Question 14 · [previous study design](#) · [open in the official exam PDF \(p. 36\)](#) ↗

Photoelectric experiment: current–voltage graphs and the E_k – f graph

a. What point P on the I–V graph represents (1 mark)

b. Sketch the graph for higher intensity (2 marks)

c. Sketch the graph for violet light at the higher intensity (2 marks)

d. Work function from the E_k – f graph (1 mark)

e. Planck's constant from the graph, in J s (2 marks)

f. A photoelectric observation the wave model can't explain (1 mark)

Question 8 (2 marks)

2022 VCAA exam · Section B · Question 15 · [previous study design](#) · [open in the official exam PDF \(p. 40\)](#) ↗

Neon energy levels and a 640 nm photon

a. Show a 640 nm photon corresponds to 1.94 eV (1 mark)

b. Draw the transition that emits this photon (1 mark)

Question 9 (7 marks)

2023 VCAA exam · Section B · Question 7 · [previous study design](#) · [open in the official exam PDF \(p. 22\)](#) ↗

Long-distance transmission line: resistance, voltage drop, lower voltage

a. Show the line resistance is $41\ \Omega$ (2 marks)

b. Will Town B get at least 480 kV? (3 marks)

c. Effect of transmitting at 50 kV instead (2 marks)

Question 10 (6 marks)

2024 VCAA exam · Section B · Question 8 · [open in the official exam PDF \(p. 28\)](#) ↗

Thundercloud and ground as charged parallel plates

a. Sketch the electric field between the cloud base and the ground (2 marks)

b. Electric field strength between cloud base and ground (2 marks)

c. Energy transferred by the lightning bolt (3 sig. figs) (2 marks)

Question 11 (6 marks)

2025 VCAA exam · Section B · Question 1 · [open in the official exam PDF \(p. 16\)](#) ↗

Ball falling from a ledge: time, speed, rebound, horizontal launch

a. Show the fall time is 0.606 s (1 mark)

b. Speed just before hitting the floor (2 marks)

c. Why the ball won't rebound to 1.80 m (energy) (1 mark)

d. Fall time when launched horizontally at 0.75 m s^{-1} (2 marks)

Question 12 (6 marks)

2025 VCAA exam · Section B · Question 3 · [open in the official exam PDF \(p. 20\)](#) ↗

Snowboarder at constant speed on a slope, then stopping

a. Draw and label the forces on the snowboarder (3 marks)

b. Show the total resistive force is 183 N (1 mark)

c. Distance from B to C where the snowboarder stops (2 marks)

Question 13 (5 marks)

2025 VCAA exam · Section B · Question 4 · [open in the official exam PDF \(p. 22\)](#) ↗

Racing car on a flat curve, then a banked track

a. Sideways friction force on the flat curve (2 marks)

b. Banking angle for no sideways friction (3 marks)

Question 14 (3 marks)

2025 VCAA exam · Section B · Question 7 · [open in the official exam PDF \(p. 27\)](#) ↗

Electron entering a magnetic field

a. Direction of the magnetic force on entry (1 mark)

b. Radius of the electron's path (3 sig. figs) (2 marks)

Question 15 (4 marks)

2025 VCAA exam · Section B · Question 15 · [open in the official exam PDF \(p. 38\)](#) ↗

Young's double slit: slit separation and a lower-frequency source

a. Slit separation from the fringe spacing (2 marks)

b. Effect of a lower-frequency source on the pattern (2 marks)

Question 16 (3 marks)

2025 VCAA exam · Section B · Question 18 · [open in the official exam PDF \(p. 42\)](#) ↗

Hydrogen atom emitting a 2.9 eV photon

a. Wavelength of the photon in nm (2 marks)

b. Draw the transition on the energy-level diagram (1 mark)

END OF PAPER

Marking guide follows

Stop here until you have finished the paper under exam conditions.

Marking guide

Section A – Multiple-choice questions

✓	Q	Ans	Why	How Victoria went
☐	1	C	Right-hand grip rule around the loop gives the field direction at the centre (into the page here). <i>Most popular wrong answer: D (15%).</i>	80% correct A 2% · B 3% · C 80% · D 15%
☐	2	A	Right-hand (slap) rule: the particle curving one way is positive, the undeflected one is neutral, and the one curving the other way is negative.	83% correct A 83% · B 2% · C 14% · D 1%
☐	3	B	Conservation of momentum: total momentum before (10 t at 3.0 m s ⁻¹) equals total after. Solving gives the second truck's mass as about 5 t.	78% correct A 4% · B 78% · C 6% · D 11%
☐	4	A	The photoelectric effect (threshold frequency, instant emission) needs photons. The others are wave behaviour.	81% correct A 81% · B 4% · C 6% · D 8%
☐	5	A	An inertial frame is not accelerating. Only the bus at constant velocity qualifies. Turning and speeding up are both accelerations.	82% correct A 82% · B 7% · C 6% · D 5%
☐	6	C	$\gamma = 1/\sqrt{1 - (2.99/3.00)^2} = 1/\sqrt{0.00666} \approx 12.3$.	85% correct A 3% · B 6% · C 85% · D 6%
☐	7	D	Uncertainty estimates the maximum likely difference between a measurement and the true value. It is not a mistake.	84% correct A 6% · B 3% · C 6% · D 84%
☐	8	C	No friction needed: $\tan \theta = \frac{v^2}{rg} = \frac{11^2}{25 \times 9.8} = 0.49$, so $\theta \approx 26^\circ$.	84% correct A 5% · B 6% · C 84% · D 5%
☐	9	A	Waves spreading around an obstacle or through a gap is diffraction.	84% correct A 84% · B 7% · C 6% · D 3%
☐	10	B	Diffraction is wave behaviour. Electrons show wave-like properties (de Broglie wavelength), so their patterns can be compared with X-rays.	78% correct A 9% · B 78% · C 6% · D 6%
☐	11	A	Midway between two identical like poles the fields cancel, so the net field is close to zero.	79% correct A 79% · B 5% · C 10% · D 6%
☐	12	C	$F = BIl = 2.0 \times 10^{-3} \times 5.0 \times 0.040 = 4.0 \times 10^{-4} \text{ N}$.	83% correct A 6% · B 8% · C 83% · D 2%
☐	13	B	For the same power, higher voltage means lower current, and line loss is I^2R , so losses fall.	80% correct A 3% · B 80% · C 7% · D 9%
☐	14	B	Precision is about how close repeated measurements are to each other (little variation), not closeness to the true value.	85% correct A 2% · B 85% · C 3% · D 10%
☐	15	C	In circular motion the acceleration points to the centre: downwards at the top and upwards at the bottom.	78% correct A 12% · B 2% · C 78% · D 9%

✓	Q	Ans	Why	How Victoria went
☐	16	A	Moving up: GPE rises and the spring shortens (less elastic PE). With no losses the total stays constant.	78% correct A 78% · B 6% · C 3% · D 12%
☐	17	B	A static field is constant in time. A non-uniform field varies from place to place across the region.	81% correct A 6% · B 81% · C 8% · D 5%
☐	18	D	Diffraction spacing depends on λ/w . Same crystal (same spacing) and the same pattern means the same wavelength.	91% correct A 3% · B 2% · C 4% · D 91%
☐	19	B	The beamline's proper length is 100 m in the lab. The protons see it contracted: $L = L_0/\gamma = 100/2.10 = 47.6 \text{ m}$.	80% correct A 2% · B 80% · C 5% · D 13%
☐	20	A	Zero error and a consistent 5% over-read are systematic (same every time). Parallax and sticking pointers vary between readings, so they are random.	78% correct A 78% · B 10% · C 5% · D 7%

Multiple-choice score: _____ / 20

Section B – Short-answer questions

Question 1 (5 marks) · state average 3.6 / 5 (72%)

2022 VCAA exam · Section B · Question 1

a. Direction of current in side KL in Position 1, with justification

- ☐ K to L
- ☐ Justified with a correctly applied, named right-hand (slap/palm) rule linking field, current and force for the given rotation

Answer / notes: K to L, by the right-hand slap rule. A vague “right-hand rule” didn’t earn the justification mark.

State average 1.2 / 2 (60%); 49% scored full marks. Mark distribution: 0: 27% · 1: 25% · 2: 49%

Your mark: _____ / 2

b. Direction of current in KL in Position 3

- ☐ L to K (the commutator has reversed it)

Answer / notes: L to K.

State average 0.7 / 1 (70%); 66% scored full marks. Mark distribution: 0: 34% · 1: 66%

Your mark: _____ / 1

c. Current in the coil from the force on KL

- ☐ $I = \frac{F}{lB}$ with substitution
- ☐ $I = 3.0 \text{ A}$

Answer / notes: $\frac{0.15}{0.10 \times 0.5} = 3.0 \text{ A}$

State average 1.7 / 2 (85%); 82% scored full marks. Mark distribution: 0: 15% · 1: 3% · 2: 82%

Your mark: _____ / 2

Question 2 (7 marks) · state average 4.9 / 7 (70%)

2022 VCAA exam · Section B · Question 3

a. Increase in kinetic energy of each ion

- ☐ $\Delta E_k = qV$
- ☐ $2.4 \times 10^{-16} \text{ J}$

Answer / notes: $1.6 \times 10^{-19} \times 1500 = 2.4 \times 10^{-16} \text{ J}$

State average 1.5 / 2 (75%); 72% scored full marks. Mark distribution: 0: 26% · 1: 3% · 2: 72%

Your mark: _____ / 2

b. Show the ion speed is $3.16 \times 10^5 \text{ m s}^{-1}$

☐ $\frac{1}{2}mv^2 = 2.4 \times 10^{-16}$

☐ $v = \sqrt{2 \times 2.4 \times 10^{-16} / 4.80 \times 10^{-27}} = 3.16 \times 10^5 \text{ m s}^{-1}$

Answer / notes: Rearrange $\text{KE} = \frac{1}{2}mv^2$.

State average 1.5 / 2 (75%); 71% scored full marks. Mark distribution: 0: 26% · 1: 4% · 2: 71%

Your mark: _____ / 2

c. Diameter of the semicircular path

☐ $r = \frac{mv}{qB}$

☐ $r \approx 0.095 \text{ m}$

☐ $D = 2r \approx 0.19 \text{ m}$

Answer / notes: $r = 0.0948 \text{ m}$, so $D = 0.19 \text{ m}$. Forgetting to double the radius was the common error.

State average 1.9 / 3 (63%); 52% scored full marks. Mark distribution: 0: 32% · 1: 4% · 2: 12% · 3: 52%

Your mark: _____ / 3

Question 3 (7 marks) · state average 4.9 / 7 (70%)

2022 VCAA exam · Section B · Question 5

a. Power produced by the generator

☐ $P = VI = 41.5 \text{ kW}$

Answer / notes: $415 \times 100 = 41.5 \text{ kW}$

State average 0.9 / 1 (90%); 83% scored full marks. Mark distribution: 0: 17% · 1: 83%

Your mark: _____ / 1

b. Can the factory get its 40 kW? Justify with calculations

☐ Line loss $= I^2 R = 100^2 \times 2.00 = 20 \text{ kW}$

☐ Power delivered $= 41.5 - 20 = 21.5 \text{ kW}$

☐ No, 21.5 kW is less than the 40 kW required

Answer / notes: No. Only 21.5 kW arrives.

State average 2.2 / 3 (73%); 65% scored full marks. Mark distribution: 0: 22% · 1: 5% · 2: 8% · 3: 65%

Your mark: _____ / 3

c. Power delivered with ideal 1:10 and 10:1 transformers

☐ Transmission current falls to 10 A (voltage 4150 V)

☐ Loss $= 10^2 \times 2.00 = 200 \text{ W}$

☐ Delivered $\approx 41.3 \text{ kW}$

Answer / notes: $41.5 - 0.2 = 41.3 \text{ kW}$

State average 1.8 / 3 (60%); 55% scored full marks. Mark distribution: 0: 34% · 1: 6% · 2: 5% · 3: 55%

Your mark: _____ / 3

Question 4 (5 marks) · state average 3.5 / 5 (70%)

2022 VCAA exam · Section B · Question 6

a. Magnetic field strength from the flux

☐ $B = \Phi / A$ (number of turns not included)

☐ $B = 0.020 \text{ T}$

Answer / notes: $1.2 \times 10^{-3} / 0.060 = 0.020 \text{ T}$

State average 1.6 / 2 (80%); 77% scored full marks. Mark distribution: 0: 21% · 1: 2% · 2: 77%

Your mark: _____ / 2

b. Average EMF over the first quarter turn at 2.5 Hz

- ☐ Quarter turn: $\Delta t = T/4 = 0.10 \text{ s}$
- ☐ $\Delta\Phi = 1.2 \times 10^{-3} \text{ Wb}$
- ☐ $\varepsilon = N \frac{\Delta\Phi}{\Delta t} = 2.4 \text{ V}$

Answer / notes: $200 \times 1.2 \times 10^{-3} / 0.10 = 2.4 \text{ V}$. Using 0.4 s was the common error.

State average 1.9 / 3 (63%); 53% scored full marks. Mark distribution: 0: 24% · 1: 21% · 2: 1% · 3: 53%

Your mark: _____ / 3

Question 5 (10 marks) · state average 6.8 / 10 (68%)

2022 VCAA exam · Section B · Question 7

a.i. Show the component of weight down the ramp is 8.3 N

- ☐ $mg \sin 25^\circ$
- ☐ $2.0 \times 9.8 \times \sin 25^\circ = 8.3 \text{ N}$

Answer / notes: Use sine, not cosine.

State average 1.6 / 2 (80%); 77% scored full marks. Mark distribution: 0: 22% · 1: 1% · 2: 77%

Your mark: _____ / 2

a.ii. Constant friction force on the ramp

- ☐ $F_{\text{net}} = ma = 2.0 \times 3.2 = 6.4 \text{ N}$
- ☐ Friction = $8.3 - 6.4 = 1.9 \text{ N}$

Answer / notes: 1.9 N

State average 1.1 / 2 (55%); 52% scored full marks. Mark distribution: 0: 47% · 1: 1% · 2: 52%

Your mark: _____ / 2

b.i. Speed of the joined trolleys, naming the principle

- ☐ Conservation of momentum stated
- ☐ $2.0 \times 4.0 = 4.0v$
- ☐ $v = 2.0 \text{ m s}^{-1}$

Answer / notes: 2.0 m s^{-1}

State average 2.0 / 3 (67%); 47% scored full marks. Mark distribution: 0: 21% · 1: 4% · 2: 28% · 3: 47%

Your mark: _____ / 3

b.ii. Elastic or inelastic? Show calculations

- ☐ KE before = $\frac{1}{2}(2.0)(4.0)^2 = 16 \text{ J}$
- ☐ KE after = $\frac{1}{2}(4.0)(2.0)^2 = 8.0 \text{ J}$
- ☐ KE decreased, so the collision is inelastic

Answer / notes: Inelastic. KE falls from 16 J to 8 J. Forgetting to square v was the common slip.

State average 2.1 / 3 (70%); 61% scored full marks. Mark distribution: 0: 21% · 1: 11% · 2: 7% · 3: 61%

Your mark: _____ / 3

Question 6 (16 marks) · state average 12.2 / 16 (76%)

2022 VCAA exam · Section B · Question 10

a. Classify controlled, dependent and independent variables

- ☐ Controlled: launch speed (or ball mass)
- ☐ Dependent: range
- ☐ Independent: launch angle

Answer / notes: Controlled: launch speed or mass. Dependent: range. Independent: launch angle.

State average 2.6 / 3 (87%); 73% scored full marks. Mark distribution: 0: 5% · 1: 9% · 2: 12% · 3: 73%

Your mark: _____ / 3

b. Plot the range–angle data with scales, uncertainty bars and a curve

- ☐ Sensible linear scales on both axes
- ☐ Axes labelled with units
- ☐ All data points plotted accurately
- ☐ Uncertainty bars of ± 1 m on at least three points
- ☐ Smooth curve of best fit (not dot-to-dot)
- ☐ Curve through or close to the uncertainty bars

Answer / notes: A smooth, symmetric-ish curve peaking near 45° at about 40 m.

State average 5.2 / 6 (87%); 51% scored full marks. Mark distribution: 0: 2% · 1: 0.5% · 2: 1% · 3: 3% · 4: 10% · 5: 33% · 6: 51%

Your mark: _____ / 6

c. Maximum range and the angle giving it, from the graph

- ☐ Maximum range $\approx 40\text{--}41$ m (from the curve)
- ☐ Angle $\approx 45^\circ$

Answer / notes: About 40.5 m at about 45° , read from your curve.

State average 1.4 / 2 (70%); 57% scored full marks. Mark distribution: 0: 16% · 1: 27% · 2: 57%

Your mark: _____ / 2

d.i. Theoretical range at 25 m s^{-1} and 30°

- ☐ $R = \frac{u^2 \sin 2\theta}{g}$ with $\sin 60^\circ$
- ☐ $R \approx 55$ m

Answer / notes: $\frac{25^2 \sin 60^\circ}{9.8} = 55.2$ m. The common error was using $\sin 30^\circ$.

State average 1.7 / 2 (85%); 81% scored full marks. Mark distribution: 0: 15% · 1: 5% · 2: 81%

Your mark: _____ / 2

d.ii. Evaluate whether air resistance can be ignored

- ☐ No, air resistance cannot be ignored
- ☐ The theoretical range (55 m) is much larger than the measured range (37 m) at 30°
- ☐ The difference is far larger than the ± 1 m uncertainty, so it is significant

Answer / notes: No. The measured range is well below theory, and the gap lies far outside the uncertainty.

State average 1.3 / 3 (43%); 4% scored full marks. Mark distribution: 0: 26% · 1: 24% · 2: 45% · 3: 4%

Your mark: _____ / 3

Question 7 (9 marks) · state average 6.5 / 9 (72%)

2022 VCAA exam · Section B · Question 14

a. What point P on the I–V graph represents

- ☐ Stopping (cut-off) voltage

Answer / notes: The stopping voltage.

State average 0.8 / 1 (80%); 73% scored full marks. Mark distribution: 0: 27% · 1: 73%

Your mark: _____ / 1

b. Sketch the graph for higher intensity

- ☐ Same stopping voltage (P unchanged)
- ☐ Higher saturation (maximum) photocurrent

Answer / notes: Same P, higher plateau.

State average 1.8 / 2 (90%); 86% scored full marks. Mark distribution: 0: 9% · 1: 5% · 2: 86%

Your mark: _____ / 2

c. Sketch the graph for violet light at the higher intensity

- ☐ Stopping voltage further from zero than P (higher-frequency light, more energetic electrons)
- ☐ Photocurrent plateau above the original

Answer / notes: Larger stopping voltage, higher plateau.

State average 1.4 / 2 (70%); 50% scored full marks. Mark distribution: 0: 17% · 1: 34% · 2: 50%

Your mark: _____ / 2

d. Work function from the E_k - f graph

☐ 3.4 eV (magnitude of the y-intercept)

Answer / notes: 3.4 eV, the y-intercept. Not from assuming a value of h .

State average 0.7 / 1 (70%); 65% scored full marks. Mark distribution: 0: 35% · 1: 65%

Your mark: _____ / 1

e. Planck's constant from the graph, in J s

☐ Gradient = rise/run = $\frac{3.4}{8.1 \times 10^{14}} = 4.2 \times 10^{-15}$ eV s

☐ Converted: $\times 1.6 \times 10^{-19} = 6.7 \times 10^{-34}$ J s

Answer / notes: 6.7×10^{-34} J s. Leaving it in eV s, or using run/rise, were the common errors.

State average 1.1 / 2 (55%); 48% scored full marks. Mark distribution: 0: 40% · 1: 12% · 2: 48%

Your mark: _____ / 2

f. A photoelectric observation the wave model can't explain

☐ Any one: a threshold frequency exists; no time delay; electron energy is independent of intensity

Answer / notes: For example, the threshold frequency.

State average 0.7 / 1 (70%); 63% scored full marks. Mark distribution: 0: 37% · 1: 63%

Your mark: _____ / 1

Question 8 (2 marks) · state average 1.4 / 2 (70%)

2022 VCAA exam · Section B · Question 15

a. Show a 640 nm photon corresponds to 1.94 eV

☐ $E = \frac{hc}{\lambda} = \frac{4.14 \times 10^{-15} \times 3.0 \times 10^8}{640 \times 10^{-9}} = 1.94$ eV

Answer / notes: Relate the wavelength to the energy. Just subtracting two levels isn't enough.

State average 0.7 / 1 (70%); 66% scored full marks. Mark distribution: 0: 34% · 1: 66%

Your mark: _____ / 1

b. Draw the transition that emits this photon

☐ Downward arrow from 21.74 eV to 19.80 eV

Answer / notes: From 21.74 eV down to 19.80 eV.

State average 0.7 / 1 (70%); 72% scored full marks. Mark distribution: 0: 28% · 1: 72%

Your mark: _____ / 1

Question 9 (7 marks) · state average 4.8 / 7 (69%)

2023 VCAA exam · Section B · Question 7

a. Show the line resistance is 41 Ω

☐ $P_{loss} = I^2 R$

☐ $R = \frac{20 \times 10^6}{700^2} \approx 41 \Omega$

Answer / notes: $20 \times 10^6 / 490\,000 = 40.8 \Omega$

State average 1.6 / 2 (80%); 78% scored full marks. Mark distribution: 0: 20% · 1: 2% · 2: 78%

Your mark: _____ / 2

b. Will Town B get at least 480 kV?

☐ Voltage drop = $IR = 700 \times 41 \approx 28.7$ kV

☐ Voltage at Town B $\approx 500 - 28.7 = 471$ kV

☐ No, 471 kV is below 480 kV

Answer / notes: No. About 471 kV arrives.

State average 2.1 / 3 (70%); 66% scored full marks. Mark distribution: 0: 25% · 1: 6% · 2: 3% · 3: 66%

Your mark: _____ / 3

c. Effect of transmitting at 50 kV instead

- ☐ For the same power the current would be 10 times larger (7000 A)
- ☐ Loss = $I^2 R$ would be 100 times larger, more than the power supplied, so the system could not work

Answer / notes: The current rises tenfold and the losses a hundredfold.

State average 1.1 / 2 (55%); 48% scored full marks. Mark distribution: 0: 42% · 1: 10% · 2: 48%

Your mark: _____ / 2

Question 10 (6 marks) · state average 4.3 / 6 (72%)

2024 VCAA exam · Section B · Question 8

a. Sketch the electric field between the cloud base and the ground

- ☐ Parallel, evenly spaced lines between cloud base and ground
- ☐ Arrows pointing upwards (from positive ground to negative cloud base)

Answer / notes: Uniform, parallel field lines pointing up.

State average 1.7 / 2 (85%); 78% scored full marks. Mark distribution: 0: 7% · 1: 14% · 2: 78%

Your mark: _____ / 2

b. Electric field strength between cloud base and ground

- ☐ $E = V/d$ with 1.20 GV converted to $1.20 \times 10^9 \text{ V}$
- ☐ $E \approx 1.41 \times 10^6 \text{ V m}^{-1}$

Answer / notes: $1.20 \times 10^9 / 850 = 1.41 \times 10^6 \text{ V m}^{-1}$

State average 1.7 / 2 (85%); 82% scored full marks. Mark distribution: 0: 15% · 1: 3% · 2: 82%

Your mark: _____ / 2

c. Energy transferred by the lightning bolt (3 sig. figs)

- ☐ $E = VIt$ with correct substitution
- ☐ $2.16 \times 10^9 \text{ J}$

Answer / notes: $1.20 \times 10^9 \times 30.0 \times 10^3 \times 60.0 \times 10^{-6} = 2.16 \times 10^9 \text{ J}$

State average 0.9 / 2 (45%); 41% scored full marks. Mark distribution: 0: 55% · 1: 5% · 2: 41%

Your mark: _____ / 2

Question 11 (6 marks) · state average 4.3 / 6 (72%)

2025 VCAA exam · Section B · Question 1

a. Show the fall time is 0.606 s

- ☐ $1.80 = \frac{1}{2}(9.8)t^2$ solved to $t = 0.606 \text{ s}$

Answer / notes: $t = \sqrt{2 \times 1.80 / 9.8} = 0.606 \text{ s}$

State average 0.8 / 1 (80%); 84% scored full marks. Mark distribution: 0: 16% · 1: 84%

Your mark: _____ / 1

b. Speed just before hitting the floor

- ☐ $v = u + at$ or $v^2 = 2gh$ with substitution
- ☐ $v \approx 5.9 \text{ m s}^{-1}$

Answer / notes: $9.8 \times 0.606 = 5.94 \text{ m s}^{-1}$

State average 1.6 / 2 (80%); 79% scored full marks. Mark distribution: 0: 19% · 1: 2% · 2: 79%

Your mark: _____ / 2

c. Why the ball won't rebound to 1.80 m (energy)

- ☐ The collision is inelastic: some kinetic energy is transformed to heat, sound and deformation, so less becomes GPE

Answer / notes: Energy is lost to heat, sound and deformation in the bounce.

State average 0.7 / 1 (70%); 68% scored full marks. Mark distribution: 0: 32% · 1: 68%

Your mark: _____ / 1

d. Fall time when launched horizontally at 0.75 m s^{-1}

- ☐ Time will be the same
- ☐ Horizontal and vertical motion are independent: initial vertical velocity is still zero, with the same height and g

Answer / notes: The same. The horizontal velocity doesn't change the vertical motion.

State average 1.2 / 2 (60%); 54% scored full marks. Mark distribution: 0: 38% · 1: 9% · 2: 54%

Your mark: _____ / 2

Question 12 (6 marks) · state average 4.2 / 6 (70%)

2025 VCAA exam · Section B · Question 3

a. Draw and label the forces on the snowboarder

- ☐ Weight (gravity) vertically down
- ☐ Normal force perpendicular to the slope
- ☐ Friction/resistance up the slope (no extra “net force” arrow)

Answer / notes: Three forces: weight, normal and friction/drag. There is no net force, since the speed is constant.

State average 2.4 / 3 (80%); 61% scored full marks. Mark distribution: 0: 5% · 1: 11% · 2: 23% · 3: 61%

Your mark: _____ / 3

b. Show the total resistive force is 183 N

- ☐ Constant speed, so resistance = $mg \sin 15^\circ = 72.0 \times 9.8 \times \sin 15^\circ \approx 183 \text{ N}$

Answer / notes: Balances the component of weight down the slope.

State average 0.8 / 1 (80%); 75% scored full marks. Mark distribution: 0: 25% · 1: 75%

Your mark: _____ / 1

c. Distance from B to C where the snowboarder stops

- ☐ Work–energy: $Fd = \frac{1}{2}mv^2$, so $183d = \frac{1}{2}(72.0)(10.0)^2 = 3600 \text{ J}$
- ☐ $d \approx 19.7 \text{ m}$

Answer / notes: $d = 3600/183 = 19.7 \text{ m}$. Using $a = 9.8$ was the common error.

State average 1.0 / 2 (50%); 46% scored full marks. Mark distribution: 0: 47% · 1: 7% · 2: 46%

Your mark: _____ / 2

Question 13 (5 marks) · state average 4.2 / 5 (84%)

2025 VCAA exam · Section B · Question 4

a. Sideways friction force on the flat curve

- ☐ $F = \frac{mv^2}{r}$ with correct substitution
- ☐ $F = 3.0 \times 10^3 \text{ N}$

Answer / notes: $\frac{800 \times 30^2}{240} = 3000 \text{ N}$

State average 1.8 / 2 (90%); 88% scored full marks. Mark distribution: 0: 10% · 1: 3% · 2: 88%

Your mark: _____ / 2

b. Banking angle for no sideways friction

- ☐ Uses $\tan \theta = \frac{v^2}{rg}$
- ☐ Correct substitution $\frac{30^2}{240 \times 9.8}$
- ☐ $\theta \approx 21^\circ$ (20.9°)

Answer / notes: $\tan \theta = 0.383$, so $\theta = 20.9^\circ$. Using sin or cos was the common error.

State average 2.4 / 3 (80%); 76% scored full marks. Mark distribution: 0: 16% · 1: 2% · 2: 5% · 3: 76%

Your mark: _____ / 3

Question 14 (3 marks) · state average 2.1 / 3 (70%)

2025 VCAA exam · Section B · Question 7

a. Direction of the magnetic force on entry

- ☐ Down the page (reversed from a positive charge because the electron is negative)

Answer / notes: Down the page. Many forgot the negative charge.

State average 0.6 / 1 (60%); 62% scored full marks. Mark distribution: 0: 38% · 1: 62%

Your mark: _____ / 1

b. Radius of the electron's path (3 sig. figs)

- ☐ $r = \frac{mv}{qB}$ with substitution
- ☐ $r \approx 1.78 \text{ m}$

Answer / notes: $\frac{9.11 \times 10^{-31} \times 7.80 \times 10^7}{1.60 \times 10^{-19} \times 2.50 \times 10^{-4}} = 1.78 \text{ m}$

State average 1.5 / 2 (75%); 64% scored full marks. Mark distribution: 0: 15% · 1: 20% · 2: 64%

Your mark: _____ / 2

Question 15 (4 marks) · state average 2.8 / 4 (70%)

2025 VCAA exam · Section B · Question 15

a. Slit separation from the fringe spacing

- ☐ $d = \frac{\lambda L}{\Delta x}$ with units in metres
- ☐ $d = 4.0 \times 10^{-4} \text{ m}$

Answer / notes: $\frac{550 \times 10^{-9} \times 2.00}{2.75 \times 10^{-3}} = 4.0 \times 10^{-4} \text{ m}$

State average 1.5 / 2 (75%); 71% scored full marks. Mark distribution: 0: 24% · 1: 5% · 2: 71%

Your mark: _____ / 2

b. Effect of a lower-frequency source on the pattern

- ☐ Lower frequency means longer wavelength
- ☐ $\Delta x = \lambda L / d$, so the bright bands move further apart

Answer / notes: The bands spread out.

State average 1.3 / 2 (65%); 59% scored full marks. Mark distribution: 0: 29% · 1: 12% · 2: 59%

Your mark: _____ / 2

Question 16 (3 marks) · state average 2.1 / 3 (70%)

2025 VCAA exam · Section B · Question 18

a. Wavelength of the photon in nm

- ☐ $\lambda = hc / E$ with substitution
- ☐ $\lambda \approx 428 \text{ nm}$

Answer / notes: $\frac{1.24 \times 10^{-6}}{2.9} = 4.28 \times 10^{-7} \text{ m} = 428 \text{ nm}$

State average 1.4 / 2 (70%); 62% scored full marks. Mark distribution: 0: 26% · 1: 12% · 2: 62%

Your mark: _____ / 2

b. Draw the transition on the energy-level diagram

- ☐ Downward arrow from $n = 5$ (13.1 eV) to $n = 2$ (10.2 eV)

Answer / notes: $n = 5$ down to $n = 2$ (a 2.9 eV gap). An upward arrow means absorption.

State average 0.7 / 1 (70%); 70% scored full marks. Mark distribution: 0: 30% · 1: 70%

Your mark: _____ / 1

Your result

Total: _____ / 121 State average on these questions: **89 / 121 (73%)**

Topics to revise:

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